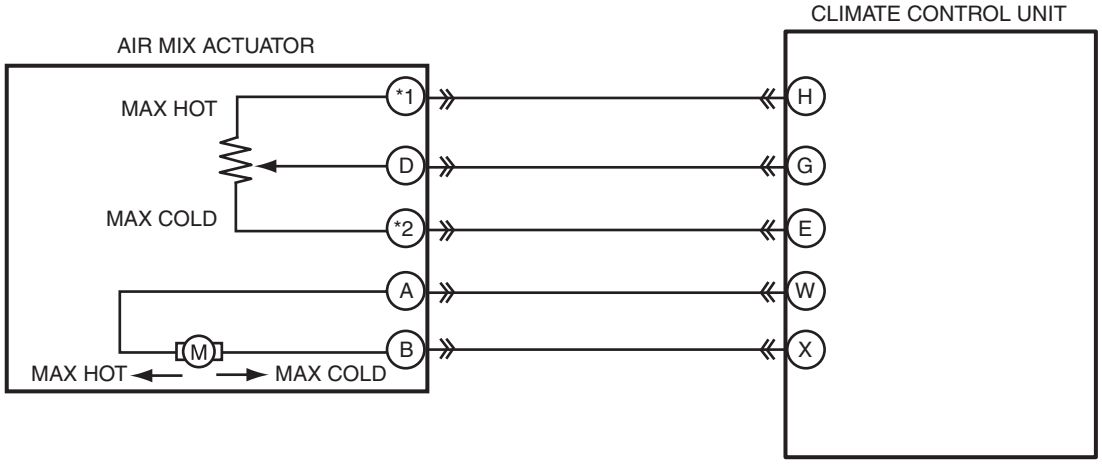
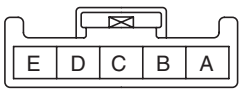
DTC B13AB:12, B13AB:13 [FULL-AUTO AIR CONDITIONER]

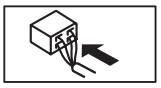
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System malfunction location	• B13AB:12: Air mix actuator (potentiometer) circuit short to power supply • B13AB:13: Air mix actuator (potentiometer) circuit open
Detection condition	• Malfunction in wiring harness between air mix actuator and climate control unit
Fail-safe function	**Malfunction determined when ignition switched ON** • Air mix actuator drive signal is stopped right when the malfunction is determined. When the set temperature is 18.0 degrees C, it is fixed at MAX COLD, and when it is 32.0 degrees C, it is fixed MAX HOT. **Malfunction already exists when ignition switched ON** • Control based on ambient temperature.When the set temperature is 18.0 degrees C, it is fixed at MAX COLD, and when it is 32.0 degrees C, it is fixed MAX HOT.
Possible cause	• Connector or terminal malfunction • Air mix actuator malfunction • Open circuit in wiring harness between climate control unit and air mix actuator • Short to power supply in wiring harness between climate control unit and air mix actuator • Climate control unit malfunction

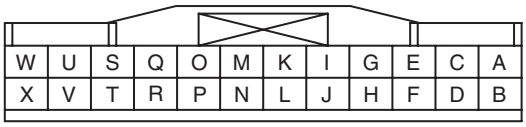


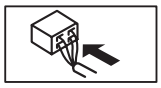
AIR MIX ACTUATOR WIRING HARNESS-SIDE CONNECTOR





CLIMATE CONTROL UNIT WIRING HARNESS-SIDE CONNECTOR





*1 : L.H.D. : C
R.H.D. : E
*2 : L.H.D. : E
R.H.D. : C

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**INSPECT AIR MIX ACTUATOR CONNECTOR** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the air mix actuator connector. (See AIR MIX ACTUATOR REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) • Inspect the connector and terminals (corrosion, damage, pin disconnection). • Are the connector and terminals normal?	Yes	Go to the next step.
		No	Repair/replace the connector or terminal. After repair procedure, go to the next step.
2	**INSPECT AIR MIX ACTUATOR** • Inspect the air mix actuator. (See AIR MIX ACTUATOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is it normal?	Yes	Go to the next step.
		No	Replace the air mix actuator. (See AIR MIX ACTUATOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].) Go to the next step.

STEP	INSPECTION		ACTION
3	**INSPECT AIR MIX ACTUATOR (POTENTIOMETER) CIRCUIT FOR OPEN CIRCUIT** • Disconnect the climate control unit connector. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) • Disconnect the air mix actuator connector. (See AIR MIX ACTUATOR REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) • Inspect for continuity between the following terminals (wiring harness-side): **L.H.D.** — Climate control unit terminal H—air mix actuator terminal C — Climate control unit terminal G—air mix actuator terminal D — Climate control unit terminal E—air mix actuator terminal E **R.H.D.** — Climate control unit terminal H—air mix actuator terminal E — Climate control unit terminal G—air mix actuator terminal D — Climate control unit terminal E—air mix actuator terminal C • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and air mix actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
4	**INSPECT AIR MIX ACTUATOR (POTENTIOMETER) CIRCUIT FOR SHORT TO POWER SUPPLY** • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at the following terminals (wiring harness-side): — Climate control unit terminal H — Climate control unit terminal G — Climate control unit terminal E • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and air mix actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to the next step.
5	**VERIFY CLIMATE CONTROL UNIT CONNECTOR CONDITION** • Inspect the connector and terminals (corrosion, damage, pin disconnection). • Are the connector and terminals normal?	Yes	Go to the next step.
		No	Repair/replace the malfunctioning vehicle wiring harness, connector, or terminal. After repair procedure, go to the next step.
6	**VERIFY THAT SAME DTC IS NOT OUTPUT AGAIN** • Reconnect the disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the past malfunction from memory. (See CLEARING DTC [FULL-AUTO AIR CONDITIONER].) • Verify DTCs. (See DTC DISPLAY [FULL-AUTO AIR CONDITIONER].) • Is DTC B13AB:12, B13AB:13 output?	Yes	Repeat the inspection from Step 1. • If the malfunction does not recur, go to the next step. • If the malfunction recurs, replace the climate control unit. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) Go to the next step.
		No	Go to the next step.
7	**VERIFY THAT NO OTHER DTCs ARE PRESENT** • Verify other DTCs displayed. (See DTC DISPLAY [FULL-AUTO AIR CONDITIONER].) • Are any other DTCs output?	Yes	Perform the corresponding DTC inspection. (See DTC TABLE [FULL-AUTO AIR CONDITIONER].)
		No	DTC troubleshooting completed.